Week 3 · Heading Control

USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering, Chungnam National University | revised 2026-09-04

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	**Control Engineering** (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	**Control System Design** (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	**Advanced Control Engineering** (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	**Sensor Signal Processing and Fusion** (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	**Capstone Design** (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

- **Course**: USV Guidance, Navigation and Control (Graduate)
- **Department**: Autonomous Vehicle System Engineering, Chungnam National University
- **This week**: ① why proportional control alone reaches the setpoint on this axis and could not on the last ② where the derivative term goes when the controlled variable is an angle ③ the wrap at $\pm 180^\circ$, and one line of arithmetic that fixes it

★ Prerequisites from the previous week

- The signal chain: **command → controller → allocation → plant → measurement**, each stage a subsystem.
- The finding of §2-4: a term must be substituted into the equation of motion before its effect is assumed. This week is that lesson applied a second time, with the opposite answer.
- From Week 1: $M_{66} = 42.65$ kg·m², $N_r = -42.65$ N·m per rad/s, and the nonlinear yaw damping $N_h = N_r(1 + 10|r|)r$.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. State why the heading loop is type 1 and predict, without computing, that proportional control leaves no steady-state error.
2. Choose K_p and K_d for a specified ω_n and ζ from the yaw equation of motion.
3. Explain why the derivative term damps this loop and did not damp the loop of Week 2, in terms of where each one lands in the equation of motion.
4. Implement the smallest-signed-angle wrap and state what a controller does without it.
5. Account for the difference between a predicted overshoot and a measured one when the hull's damping is not linear.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b, with Simulink
Toolboxes	none beyond Simulink for this week
MSS	`Tools/MSS`, added by the setup script
Course folder	`GradCourse/lectures/W03_simulink`
Expected duration	60 min theory, 75 min laboratory

Part 1 · Theory

3-1. The yaw axis, and one structural difference

- The heading is not a velocity. It is the **integral** of one:

$$\dot{\psi} = r.$$

Deriving the second-order heading equation

- This equation is the **third row** of the 3-DOF model of §1-7, and the derivation matters because the terms dropped here are not zero, unlike those dropped in §2-1.
- Start again from $\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}$ with $\boldsymbol{\nu} = [u\ v\ r]^\top$. The yaw row is

$$\underbrace{(mx_g - N_{\dot v})\,\dot v}_{\text{sway-yaw coupling}} + (I_z - N_{\dot r})\,\dot r + \underbrace{mx_g ur + (X_{\dot u} - Y_{\dot v})\,uv}_{\text{Coriolis}} = N + N_r r + N_v v$$

- Three terms stand between this and the scalar plant. Unlike §2-1, **none of them is identically zero**, and each is dropped for a stated reason:

Term	Why it is dropped	When it bites
$(mx_g - N_{\dot v})\dot v$	sway acceleration is small once the turn has settled	during the first seconds of a turn
$(X_{\dot u} - Y_{\dot v})\,uv$	proportional to uv; v is a few per cent of u	at speed, in a hard turn
$N_v v$	the Otter has no linear sway damping, $Y_v = 0$	never, for this hull

- Dropping them, writing M_{66} for the $(6,6)$ entry of $\mathbf{M}$ and substituting $r = \dot\psi$ from the kinematics:

$$M_{66}\,\dot r = N + N_r r$$
$$M_{66}\,\ddot\psi = \tau_N + N_r\,\dot\psi$$

> **⚠ This reduction is an approximation, and Week 1 already measured the error**
>
> In §2-1 the dropped terms were exactly zero and the reduction was exact. Here they are not. Week 1 measured $v = \pm 0.1264$ m/s in a turn — real sway, produced by the very Coriolis coupling dropped above. The linear model below is therefore a **design model**, and §3-5 shows where the vessel stops obeying it.

> **✕ M_{66} is not $I_z - N_{\dot r}$**
>
> The textbook symbol suggests it is, and for a vessel whose origin sits at its centre of gravity it would be. The Otter's does not: `otter.m` places the body origin on the waterline and carries $x_g = 0.153$ m, so building $\mathbf{M}_{RB}$ transfers the inertia from CG to CO and M_{66} picks up rigid-body terms beyond I_z. Week 1 §1-12 works the number through. Take $M_{66} = 42.65$ kg·m² from the model, not from the symbol.

$$M_{66}\,\ddot\psi = \tau_N + N_r\,\dot\psi.$$

Symbol	Quantity	Value / source
M_{66}	yaw inertia including added mass	42.65 kg·m², `otter.m`
N_r	linear yaw damping	-42.65 N·m per rad/s, $-M_{66}/T_{\text{yaw}}$
τ_N	commanded yaw moment	the controller's output

- Taking Laplace transforms:

$$\frac{\psi(s)}{\tau_N(s)} = \frac{1}{s\,(M_{66}s + |N_r|)}.$$

> **★ There is a free $1/s$, and it changes everything**
>
> Week 2's plant had no integrator, so the loop was **type 0** and proportional control could never reach the setpoint. This plant has one, so the loop is **type 1** and proportional control reaches the setpoint at every gain. Nothing was tuned to achieve that. It is a property of the axis.

- The reason is the same physical argument as Week 2, run backwards. Holding a steady **speed** needs a steady force, because damping never stops. Holding a steady **heading** needs no moment at all, because a vessel that is not turning has no yaw damping to overcome. A proportional controller can supply zero moment at zero error, and that is exactly what is required.

	Week 2, surge	Week 3, heading
controlled variable	a velocity	an angle
plant	$K_u/(\tau_u s + 1)$	$1/[s(M_{66}s + \lvert N_r\rvert)]$
type	0	1
steady state needs	a non-zero force	no moment
P alone reaches the setpoint	no, at any gain	yes, at every gain

The same plant under its usual name: Nomoto

- The model just derived has a name. Every paper on ship steering writes it as **Nomoto's model** (Nomoto et al., 1957), and a student who does not recognise it will not recognise most of the steering literature.
- Keeping the sway–yaw coupling that §3-1 discarded and eliminating v between the two rows gives a **second-order** relation between rudder angle δ and yaw rate:

$$T_1T_2\,\ddot{r} + (T_1 + T_2)\,\dot{r} + r = K\left(\delta + T_3\,\dot{\delta}\right)$$

- The sway dynamics are fast compared with the yaw dynamics, so T_2 and T_3 nearly cancel. Dropping them leaves the **first-order Nomoto model**, which is what almost every autopilot is designed on:

$$\boxed{T\,\dot{r} + r = K\,\delta}, \qquad T = T_1 + T_2 - T_3$$

Symbol	Meaning
K	rudder gain — how much steady yaw rate one degree of rudder buys
T	time constant — how long the rate takes to build
K/T	turning ability; large K/T is a nimble vessel

- The Otter has no rudder. Its yaw moment comes from the **difference between two propellers**, so δ is replaced by τ_N and the same first-order form appears directly from §3-1:

$$M_{66}\,\dot{r} + |N_r|\,r = \tau_N \qquad \Longleftrightarrow \qquad \underbrace{\frac{M_{66}}{|N_r|}}_{T}\,\dot{r} + r = \underbrace{\frac{1}{|N_r|}}_{K}\,\tau_N$$

- Putting in the two numbers from `otter.m`:

$$T = \frac{42.6515}{42.6515} = 1.000000\ \mathrm{s}, \qquad K = \frac{1}{42.6515} = 0.023446\ \frac{\mathrm{rad/s}}{\mathrm{N\cdot m}}$$

- Both numbers, and every other coefficient quoted in this course, are checked against the source rather than remembered:

```
verify_constants
```

- That script rebuilds $\mathbf{M}$ exactly as `otter.m` lines 55–120 build it and compares all sixteen quoted values. It reports `every quoted value agrees with otter.m to its printed precision`.

> **i $T = 1$ s exactly, and it is not a coincidence**
>
> `otter.m` sets the yaw damping as $N_r = -M_{66}/T_{\text{yaw}}$ with $T_{\text{yaw}} = 1$ s — the damping is *defined* from a chosen time constant rather than measured. So the Nomoto T of this vessel is 1 s by construction. Read the source before quoting a hydrodynamic coefficient as if it were a measurement.

- Adding the kinematics $\dot{\psi} = r$ turns the first-order Nomoto model into the type 1 plant used for the rest of this week:

$$\frac{\psi(s)}{\tau_N(s)} = \frac{K}{s\,(Ts+1)} = \frac{1}{s\,(M_{66}s + |N_r|)}$$

- The two right-hand sides are the same expression, divided top and bottom by $|N_r|$. Nothing new has been introduced; the Nomoto form simply names the two numbers that matter.

> **⚠ Nomoto is linear, and this hull is not**
>
> Nomoto's model has a constant T. Section 3-5 shows that the Otter's damping grows with $|r|$, so its effective T **shrinks as the turn gets harder**. The nonlinear extension that repairs this is Norrbin's, $T\dot{r} + r + \alpha r^3 = K\delta$, and it is the reason a large turn overshoots less than a small one.

3-2. The control law

- The controller is proportional on the heading error and derivative on the **yaw rate**:

$$\boxed{\tau_N = K_p\ \mathrm{ssa}(\psi_d - \psi) - K_d\, r}$$

- Two choices are being made, and both are deliberate.

Choice	Reason
ssa on the error	angles wrap; §3-5 is what happens without it
$-K_d r$, not $-K_d\dot{e}$	r is measured directly by the gyro. Differentiating the error would also differentiate every step in ψ_d

3-3. Where the derivative term goes this time

- Substitute the control law into the equation of motion, ignoring the wrap for a moment:

$$M_{66}\ddot{\psi} = K_p(\psi_d - \psi) - K_d\dot{\psi} + N_r\dot{\psi}$$

$$M_{66}\ddot{\psi} + (|N_r| + K_d)\dot{\psi} + K_p\psi = K_p\psi_d.$$

- Comparing with $\ddot{\psi} + 2\zeta\omega_n\dot{\psi} + \omega_n^2\psi = \omega_n^2\psi_d$:

$$\omega_n = \sqrt{\frac{K_p}{M_{66}}}, \qquad \zeta = \frac{|N_r| + K_d}{2\sqrt{K_p M_{66}}}.$$

- Inverting for a chosen pair:

$$K_p = \omega_n^2 M_{66}, \qquad K_d = 2\zeta\sqrt{K_p M_{66}} - |N_r|.$$

- For $\omega_n = 1.53$ rad/s and $\zeta = 0.9$ this gives $K_p = 100.00$ and $K_d = 74.90$.

> **★ The same term, the opposite effect**
>
> In Week 2 the controlled variable was a velocity, so K_d multiplied an **acceleration** and landed beside the mass: $(M_{11} + K_d)\dot{u}$. Here the controlled variable is an angle, so K_d multiplies a **rate** and lands beside the damping: $(|N_r| + K_d)\dot{\psi}$. Nothing about the controller changed. The axis did.

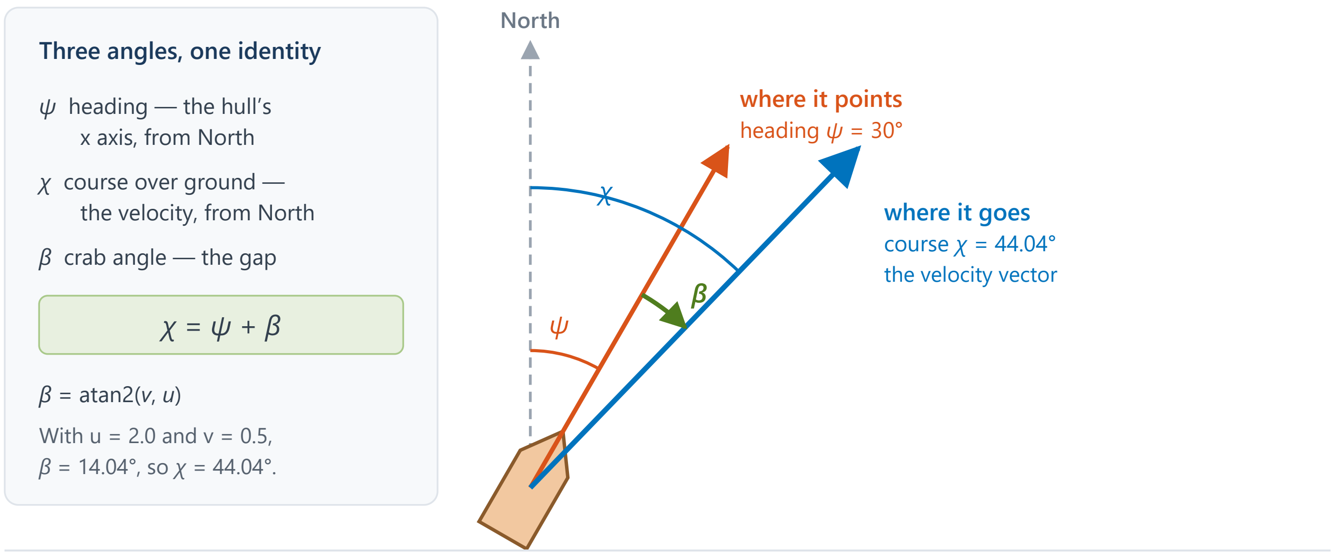
- Two further remarks, both measurable in Part 2.
  - There is **no zero** in the closed loop. The numerator is K_p alone, so the tabulated relation $M_p = \exp\left(-\pi\zeta/\sqrt{1-\zeta^2}\right)$ applies here in a way it did not in Week 2.
  - The hull already supplies damping. At $K_p = 100$ and $K_d = 0$, $\zeta = 42.65/(2\sqrt{100 \times 42.65}) = 0.327$. Most of what damps a heading loop on this vessel is N_r, not the controller.

3-4. Heading is not course — the crab angle

- A heading controller regulates where the vessel **points**. It does not regulate where the vessel **goes**, and those are two different directions.

Heading, course and the crab angle

Where the vessel POINTS and where it GOES are two different directions. A heading controller regulates the first one only.



β is non-zero whenever the sway velocity v is — in a turn, in a current, in a beam wind. It is not a disturbance to be rejected; it is what the hull is doing. Steering ψ while travelling along χ is the permanent path error that Week 5 has to remove.

Reading the figure

Element	Meaning
orange ray	the heading ψ — the direction x_b points
blue ray	the course χ — the direction the velocity actually goes
green arc	the crab angle β, the gap between them

Symbol	Definition	Unit
ψ	heading, from North to x_b	rad
χ	course over ground, from North to the velocity vector	rad
β	crab angle	rad

$$\boxed{\beta = \operatorname{atan2}(v,\ u), \qquad \chi = \psi + \beta}$$

- With the example of Week 1, $u = 2.0$ m/s and $v = 0.5$ m/s at $\psi = 30^\circ$:

$$\beta = \operatorname{atan2}(0.5,\ 2.0) = 14.04°, \qquad \chi = 44.04°.$$

- And the check: $\operatorname{atan2}(\dot{E}, \dot{N}) = \operatorname{atan2}(1.4330,\ 1.4821) = 44.04^\circ$. The course computed from the body velocity and the course computed from the NED velocity are the same number, as they must be.

> **★ β is not a disturbance**
>
> The crab angle is non-zero whenever $v \neq 0$ — in a turn, in a current, in a beam wind. It is what the hull is doing, not an error to be removed. Week 1 measured $\beta = -20.3^\circ$ in the turning run of a vessel with **no** sway actuation at all.

> **i Two definitions of β are in circulation**
>
> Fossen writes $\beta = \arcsin(v/U)$ with $U = \sqrt{u^2 + v^2}$; the form used here is $\operatorname{atan2}(v, u)$. For $u > 0$ they are identical — at $u = 2.0$, $v = 0.5$ both give 14.036°. They part company going astern: at $u = -1.0$, $v = 0.5$ the arcsin form gives 26.6° and `atan2` gives 153.4°. Only the second is the direction the vessel is actually travelling, so this course uses `atan2` throughout.

- This week regulates ψ and lets χ fall where it may. **Week 4 cannot**: a path-following law that steers the heading onto a track while the vessel travels along the course leaves a permanent cross-track error of the order of β times the look-ahead distance.

3-5. The wrap

- Headings are angles, and angles are not numbers on a line. Commanding -170° while the vessel heads $+170^\circ$ gives

$$\psi_d - \psi = -170^\circ - 170^\circ = -340^\circ,$$

- which is a perfectly valid subtraction and a perfectly wrong error. The two headings are 20° apart.
- The **smallest signed angle** maps any angle into $(-\pi, \pi]$:

$$\operatorname{ssa}(a) = \operatorname{atan2}(\sin a,\ \cos a).$$

- Applied to the example, $\operatorname{ssa}(-340^\circ) = +20^\circ$, and the vessel turns the short way.

> **⚠ Wrap the error, not the heading**
>
> ψ itself is left unwrapped throughout this course, exactly as `otter.m` produces it. Wrapping the state would put a 360° jump into a signal that is differentiated and plotted. The wrap belongs at the **one place an angle is subtracted from an angle**, which is inside the `heading error` block and nowhere else.

3-6. Allocation, now with two demands

- The controller asks for a yaw moment. A constant forward force X_{ff} is added so that the vessel travels while it turns and the track is worth looking at. Two demands, two propellers:

$$X = T_1 + T_2, \qquad N = y_{\mathrm{pont}}\,(T_1 - T_2)$$
$$\Longrightarrow \quad T_1 = \frac{X}{2} + \frac{N}{2y_{\mathrm{pont}}}, \qquad T_2 = \frac{X}{2} - \frac{N}{2y_{\mathrm{pont}}}.$$

- The map is **square** in the (X, N) plane, so the inverse exists and is unique. Nothing is optimised and nothing is chosen. Week 5 meets the case where there are more thrusters than demands.
- Each thrust is then converted to a shaft speed by inverting the propeller curve one propeller at a time, $n = \mathrm{sign}(T)\sqrt{|T|/k}$, and saturated.

> **✎ Allocating thrust rather than shaft speed pays a dividend**
>
> Appendix A1 derived, by hand, that a pure turn needs $n_2 = -n_1\sqrt{k_{\mathrm{pos}}/k_{\mathrm{neg}}}$ rather than $-n_1$. This allocation produces that pair **by itself**: a pure yaw moment gives $T_2 = -T_1$, and because $k_{\mathrm{neg}} < k_{\mathrm{pos}}$ the reverse shaft comes out faster. Doing the arithmetic in the right units removes a whole class of error.

Part 2 · Laboratory

A. Setting up (10 min)

```
cd GradCourse/lectures/W03_simulink
W03_0_setup
```

Expected output:

```
  W03 setup complete
    plant           M66 = 42.65 kg m^2,  Nr = -42.65
    controller      Kp = 100, Kd = 74.9, ssa = 1
    closed loop     wn = 1.5312 rad/s,  zeta = 0.9000
    hull alone      zeta = 0.3265  (Kd = 0)
    reference       60 -> 60 deg at t = 5 s
    forward force   60 N
    simulation      40 s at h = 0.02 s
```

- `W03_0_setup.m` is **the only file to edit this week**. To restore a broken model: `W03_1_build_heading`.

The files of this week, in the order the sections use them

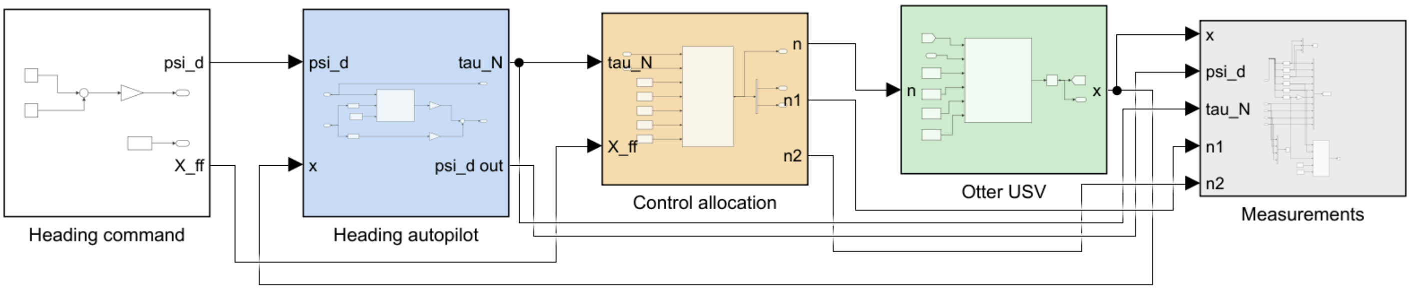
Order	File	Section	What it produces
0	`W03_0_setup.m`	A	the base workspace, so the model can be run from Simulink
1	`W03_1_build_heading.m`	B	`W03_heading_control.slx` and `img/W03_heading_control.png`
C	`W03_C_proportional_only.m`	C	`img/W03_result_P.png`
D	`W03_D_derivative_action.m`	D	`img/W03_result_D.png`
E	`W03_E_the_wrap.m`	E	`img/W03_result_ssa.png`
F	`W03_F_big_turns_overshoot_less.m`	F	`img/W03_result_size.png`

- Each laboratory section is one script. Running a section leaves exactly the numbers and the one figure that section discusses, so a class can work through the week a page at a time.
- Sections C to F build the model themselves if it is missing, so any one of them can be run first.

B. Reading the model (10 min)

i To produce this figure

`W03_1_build_heading` writes `W03_simulink/img/W03_heading_control.png` at the end of the build. The diagram belongs to the builder and to nothing else, so it changes when the model changes and not when a gain changes.



WEEK 3 - HEADING CONTROL

The same chain as Week 2 with one axis changed, and every conclusion of Week 2 reversed by that one change.

The heading is the INTEGRAL of the yaw rate, psi = int r, so the plant carries a free integrator and the loop is TYPE 1:

M66 psi_ddot = tau_N + Nr psi_dot,   M66 = 42.65, Nr = -42.65.

FOUR THINGS TO TRY

1 PROPORTIONAL ONLY. Kd = 0. The steady-state error is ZERO, at every gain. Week 2 could not do this at any gain. Nothing was tuned; the plant simply has an integrator that the surge axis did not have.

2 ADD THE DERIVATIVE. Kd acts on the YAW RATE r, which is the derivative of the controlled variable. Here it is a DAMPER:

zeta = (|Nr| + Kd) / (2 sqrt(Kp M66)).

In Week 2 the same term was a MASS and made things worse. Substitute the control law into the equation of motion before assuming.

3 THE WRAP. psi_1 = 170, psi_2 = -170, t_dn = 20. With use_ssa = 1 the vessel turns 20 deg. With use_ssa = 0 it turns 340 deg the other way, to reach the same heading.

4 BIG TURNS OVERSHOOT LESS. Kd = 0, psi_1 = 5, 20, 60, 120. The yaw damping in otter.m is Nh = Nr (1 + 10|r|) r, so a fast turn is damped harder than a slow one. No linear model can do this.

Reading the figure

Element	Meaning
`Heading command` (white)	two steps in degrees, converted to radians once; and the constant forward force
`Heading autopilot` (blue)	selectors for ψ and r, the wrapped error, and K_p, K_d
`Control allocation` (sand)	$(\tau_N, X_{\text{ff}}) \to$ two shaft speeds, by the inverse of §3-6
`Otter USV` (green)	`otter.m`, called unchanged
`Measurements` (grey)	selectors, scope, workspace log and live view

- The chain is the same one as Week 2 with the controller replaced. **One signal travels backwards**: the state $\mathbf{x}$, from which the autopilot selects ψ and r.
- Angles are in **degrees at the command and in the plots, and in radians everywhere between**. The single `deg2rad` gain inside `Heading command` is the boundary, and it is the only place a conversion happens on the way in.

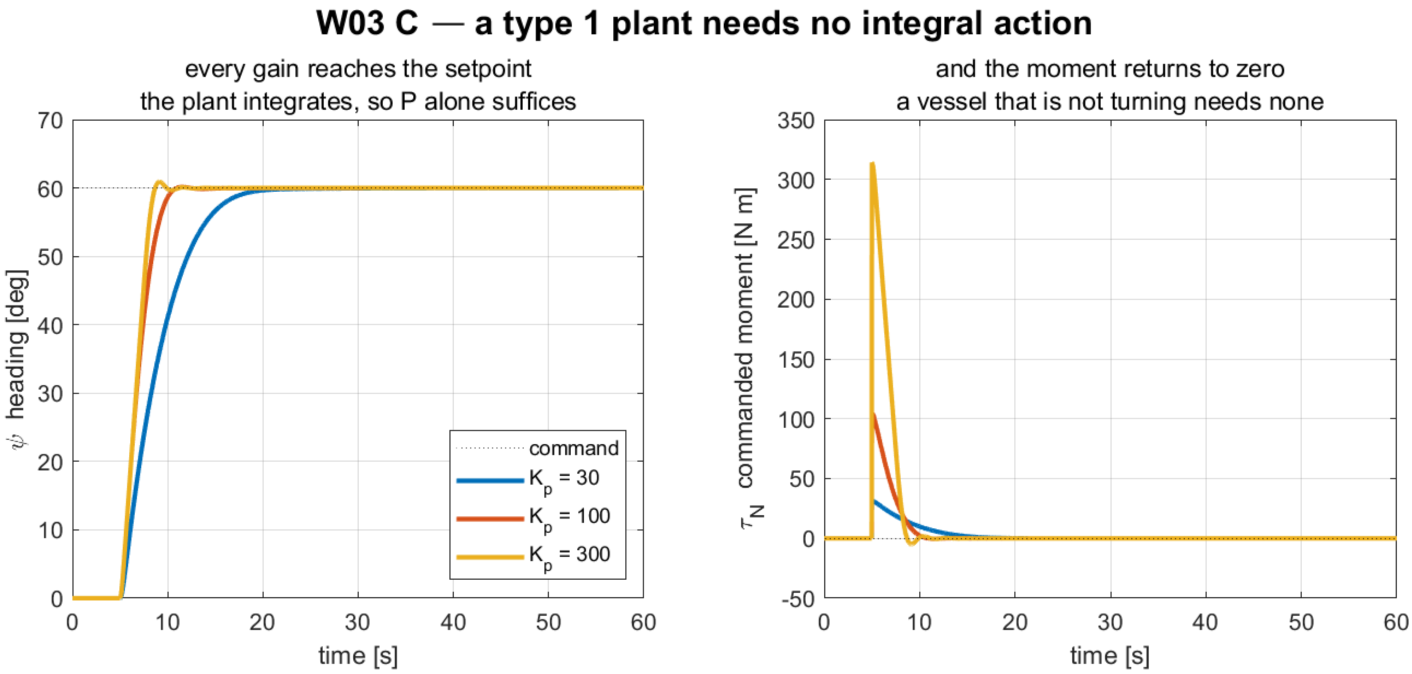
C. Proportional only (15 min)

```
W03_C_proportional_only
```

> **i To produce this figure**
>
> `W03_C_proportional_only.m` runs `W03_heading_control.slx` three times with `Kd = 0`, prints the table below, and writes `img/W03_result_P.png`.

$K_d = 0$, step to 60°:

K_p	ω_n	ζ	steady error [deg]	overshoot [%]	t_s (2%) [s]
30	0.8387	0.5962	7.9×10^{-3}	-0.01	12.24
100	1.5312	0.3265	1.8×10^{-3}	0.24	5.06
300	2.6522	0.1885	5.7×10^{-4}	1.53	3.46



Reading the figure

Element	Meaning
left panel	three step responses; all three arrive at the dashed setpoint
right panel	the commanded yaw moment τ_N, which returns to zero in every run

- The steady-state error is **zero at every gain**, to the tolerance of the solver. Week 2's table at the same place read 43.68, 13.43 and 3.73 per cent.
- The difference between the two weeks is one structural fact about the axis. No better controller was written.

What the figure says

- **Meaning.** The left panel is what the vessel did; the right panel is what the controller had to ask for to make it do that. Reading them together is what separates this week from the last.
- **Trend, in numbers.** All three headings converge on 60° and stay there. Raising K_p from 30 to 300 cuts the 2 % settling time from 12.24 to 3.46 s and buys that speed with moment: the peak demand rises from about 30 to 313 N·m, a factor of ten. Overshoot grows too, from -0.01% to 1.53%, but stays small because the hull's own damping is large.
- **Principle.** $\psi = \int r$, so the plant contains a free integrator and the loop is **type 1**. The final-value theorem then gives zero steady-state error for a step at any finite gain. **Nothing was tuned to achieve this, and no integral term was added.**

- **Why the right panel is the proof.** In steady state τ_N returns to **zero** in all three runs. A vessel that has stopped turning needs no moment to keep its heading, so the controller can be at its setpoint and demanding nothing at the same time. Week 2 could not do this: holding a speed needs a permanent force, that force can only come from a non-zero error, and the error therefore never vanished.
- **What changes with the situation.** Raising K_p trades peak actuator demand for speed and never touches the steady state. That is a different bargain from Week 2, where raising the gain bought accuracy the loop could not otherwise have. The limit here is the actuator, and section D adds the term that lets K_p rise without the response ringing.

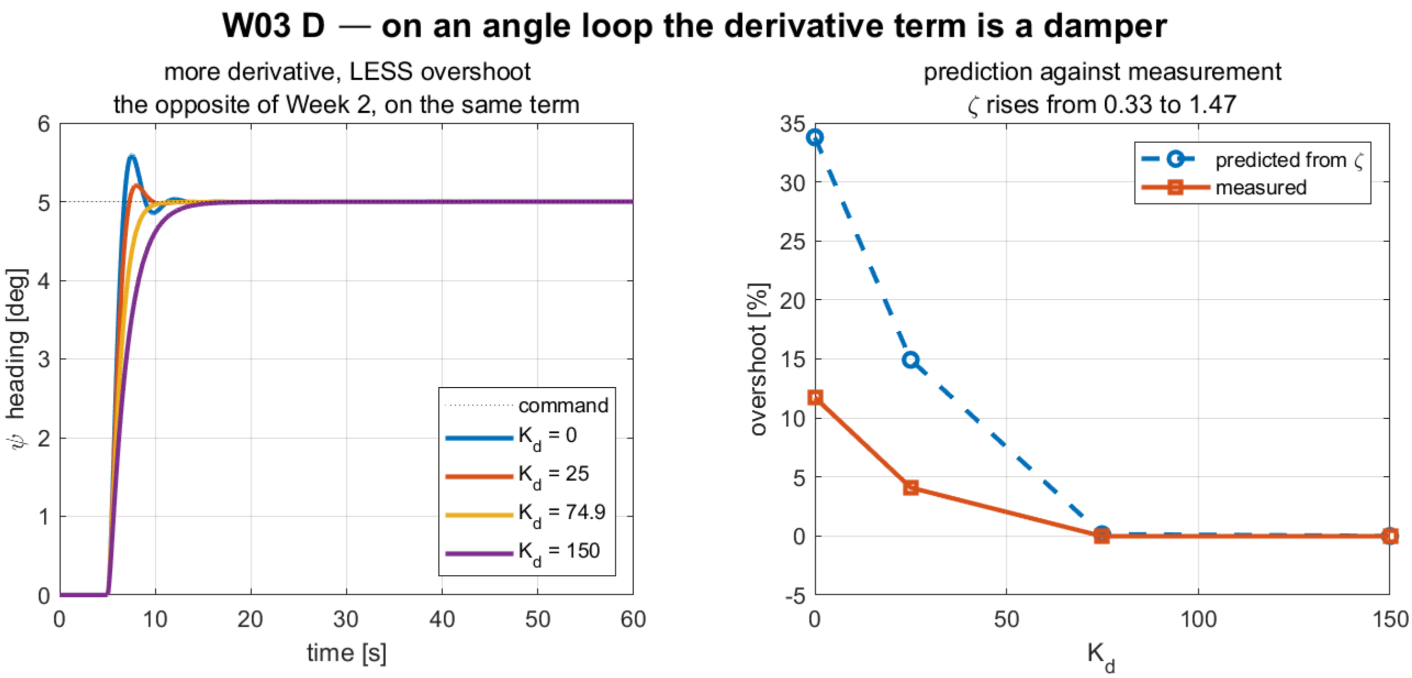
D. Derivative action (20 min)

```
W03_D_derivative_action
```

> **i To produce this figure**
>
> `W03_D_derivative_action.m` runs `W03_heading_control.slx` four times with `Kp` held and `Kd` swept, prints the table below, and writes `img/W03_result_D.png` .

$K_p = 100$ held, and a deliberately **small** step of 5°:

K_d	ζ	predicted M_p [%]	measured M_p [%]	t_s (2%) [s]
0	0.3265	33.78	11.74	5.40
25	0.5179	14.92	4.10	3.88
74.90	0.9000	0.15	-0.02	4.14
150	1.4750	0.00	-0.02	7.52



Reading the figure

Element	Meaning
left panel	four step responses; overshoot falls monotonically with K_d
right panel, dashed	overshoot predicted from ζ by the second-order formula
right panel, solid	overshoot actually measured, for the same four gains

- The **trend** is exactly as predicted: more K_d, more damping, less overshoot. Compare Week 2's table, where more K_d meant *more* overshoot.
- The **magnitudes** are not. The measured overshoot is roughly a third of the predicted one at $K_d = 0$. Section F explains why, and the explanation is not a modelling error.

What the figure says

- **Meaning.** The left panel shows the same 5° command answered by four controllers that differ in one number. The right panel puts the design formula and the simulation side by side, so the reader can see where the paper agrees with the vessel and where it does not.
- **Trend, in numbers.** Overshoot falls from 11.74% at $K_d = 0$ to zero at $K_d = 74.9$, and beyond that nothing more is bought: $K_d = 150$ also overshoots nothing but takes 7.52 s to settle instead of 4.14. **The best settling time in the table is at $K_d = 25$, which still overshoots 4.1%** — the fastest gain and the smoothest gain are not the same gain.
- **Principle.** Substituting the control law into the equation of motion gives $M_{66}\ddot{\psi} + (|N_r| + K_d)\dot{\psi} + K_p\psi = K_p\psi_d$. The derivative gain lands **beside the damping**. In Week 2 the controlled variable was a velocity, its derivative was an acceleration, and the identical term landed beside the *mass* and made the response worse. The term did not change; the axis did.
- **Why the two curves in the right panel differ.** They agree at the right-hand end and diverge at the left. The prediction is a linear result computed from N_r alone, but the hull's real damping is $N_h = N_r(1 + 10|r|)r$, which is larger whenever the vessel turns quickly. At $K_d = 0$ the response is quick, the extra damping is largest, and the true overshoot is a third of the predicted one. As K_d grows the motion slows, $|r|$ falls, and the linear prediction becomes correct.
- **What changes with the situation.** The step is only 5° for this reason. Section F holds the gains fixed and varies the step size instead, turning the same nonlinearity from a nuisance into the subject.

> **i Why the step is only 5 degrees here**
>
> The design equations of §3-3 use N_r alone. The hull's actual damping is $N_h = N_r(1 + 10|r|)r$, which is larger whenever the vessel is turning quickly. A small step keeps $|r|$ small and the linear prediction close. Section F makes the same nonlinearity the subject rather than a nuisance.

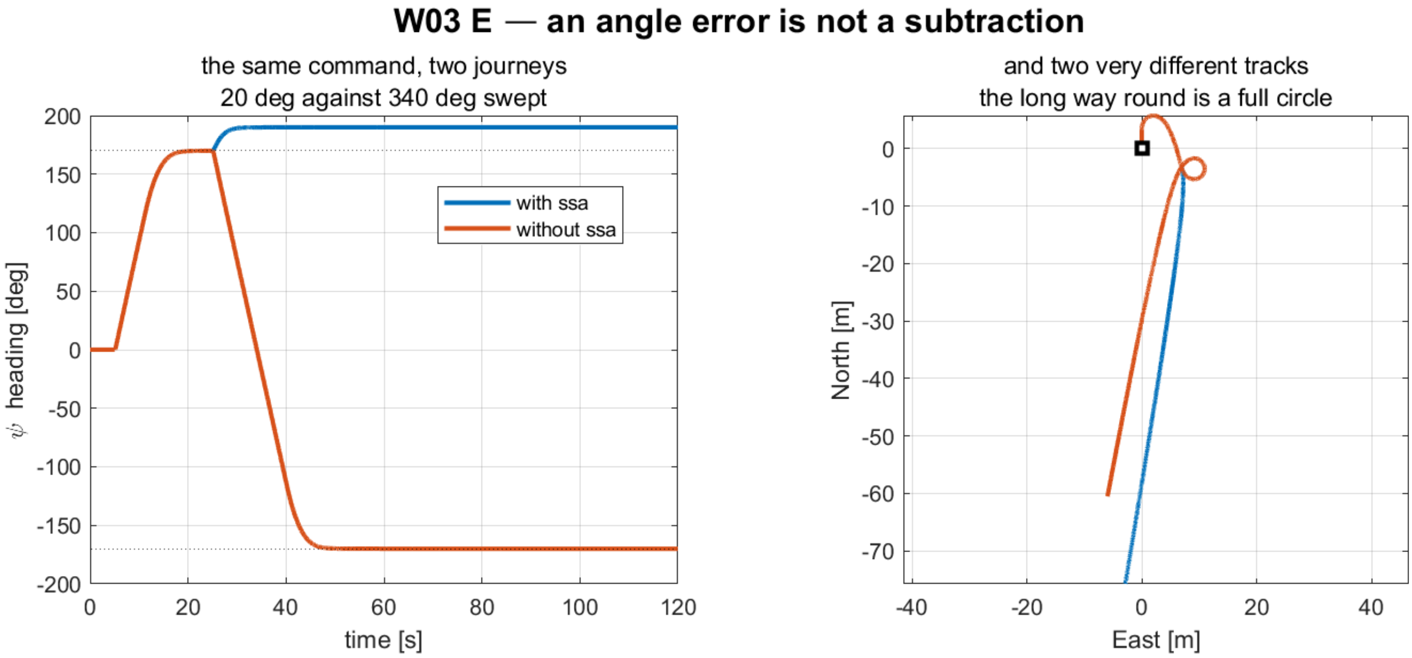
E. The wrap (15 min)

```
W03_E_the_wrap
```

> **i To produce this figure**
>
> `W03_E_the_wrap.m` runs `W03_heading_control.slx` twice — once with `use_ssa = 1`, once with `use_ssa = 0` — prints the table below, and writes `img/W03_result_ssa.png`.

The vessel is commanded to $+170^\circ$, allowed to settle, and then commanded to -170° — a change of 20°.

`use_ssa`	peak $|r|$ [deg/s]	total turn [deg]	settled ψ [deg]
1	8.375	20.0	190.00
0	19.640	340.0	-170.00



Reading the figure

Element	Meaning
left panel	the heading over time; the two dotted lines are $+170^\circ$ and -170°
left panel, blue	with `ssa` — the heading rises through 170° and stops at 190°
left panel, orange	without `ssa` — the heading falls all the way to -170°
right panel	the two tracks over the ground, from the same start marker

- Both vessels end pointing the same way. One turned 20° and the other turned 340° the other way.
- The settled headings read 190° and -170° because ψ is never wrapped in this course. They are the same heading, expressed differently, which is precisely the point.

What the figure says

- **Meaning.** Two runs of one model with one flag changed. The left panel is what each vessel believed it had to do; the right panel is what that cost in the water.
- **Trend, in numbers.** Up to $t = 25$ s the two curves are identical — both reach $+170^\circ$ and hold it. At the second command they separate completely. The blue heading climbs 20° and stops; the orange one falls 340°, through zero, all the way to -170°. Peak yaw rate more than doubles, from 8.375 to 19.640 deg/s. On the right the blue track is a gentle bend while the orange one **closes a full loop** and ends 60 m away, pointing the same way as the vessel that never left the line.
- **Principle.** $\psi_d - \psi = -170 - 170 = -340^\circ$ is a perfectly valid number and a perfectly wrong error. The smallest-signed-angle map $\mathrm{ssa}(a) = \mathrm{atan\,2}(\sin a, \cos a)$ folds it into $(-\pi, \pi]$ and returns $+20^\circ$. **The controller is identical in both runs; only the arithmetic that forms its input differs.**
- **What separates the two.** Not stability, not tuning, not the plant. Both loops are stable and both reach their setpoint. The failure is that one of them reaches it **the long way round**, and no gain adjustment would have prevented it.
- **What changes with the situation.** The two runs are indistinguishable anywhere away from the $\pm180^\circ$ boundary, which is what makes the bug dangerous: it passes every test that does not cross the boundary, and a real mission

crosses it routinely.

> ★ **One line of arithmetic separates the two runs**
>
> The only difference is `e = atan2(sin(e), cos(e))`. It costs nothing, it is one line, and without it a heading controller is wrong near a boundary it will certainly cross during a mission.

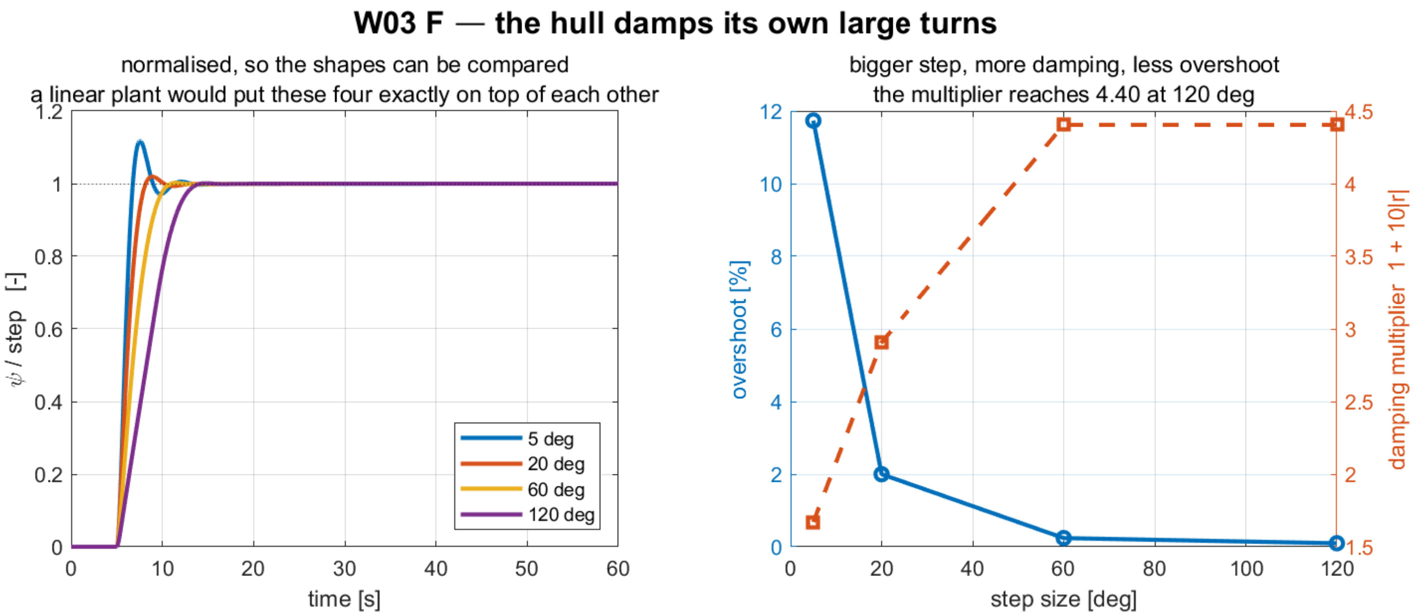
F. Big turns overshoot less (15 min)

```
W03_F_big_turns_overshoot_less
```

> **i To produce this figure**
>
> `W03_F_big_turns_overshoot_less.m` runs `W03_heading_control.slx` four times with the gains held and the step size swept, prints the table below, and writes `img/W03_result_size.png`.

$K_p = 100$, $K_d = 0$, four step sizes:

step [deg]	peak $\lvert r \rvert$ [deg/s]	overshoot [%]	t_s (2%) [s]
5	3.840	11.74	5.40
20	10.926	2.00	3.90
60	19.505	0.24	5.06
120	19.505	0.10	7.86



Reading the figure

Element	Meaning
left panel	the same steps normalised to 100%; a linear system would superimpose exactly
right panel, left axis	the measured overshoot, falling with step size
right panel, right axis	the damping multiplier $1 + 10\lvert r \rvert$ at the peak rate of each run

- A **larger** step overshoots **less**. No linear model can produce this, and normalising the responses makes it obvious: they do not superimpose.
- The mechanism is the yaw damping of `otter.m`:

$$N_h = N_r\,(1 + 10|r|)r.$$

- At the peak rate of the 120° step, 19.505 deg/s $= 0.3404$ rad/s, the damping is 4.40 times its small-signal value. The design equations of §3-3 are therefore a **small-signal** result: accurate for the 5° step and conservative for the large ones.
- The peak rate is the same for the 60° and 120° steps. Both saturate the propellers, so beyond a certain step size the vessel turns as fast as it can and no faster.

What the figure says

- **Meaning.** Four commands of different size, answered by one unchanged controller. The left panel removes the size difference by normalising, so the only thing left to compare is the **shape** of the response. The right panel puts the measured overshoot against the physical quantity that explains it.
- **Trend, in numbers.** Overshoot falls from 11.74% at 5° to 0.10% at 120° — a factor of 117, from a controller that was never retuned. Over the same range the damping multiplier $1 + 10|r|$ climbs from 1.67 to 4.40. The two curves in the right panel are mirror images, which is the claim being made: **the extra damping is the reason for the missing overshoot.**
- **Principle.** `otter.m` models yaw damping as $N_h = N_r(1 + 10|r|)r$. Damping therefore grows with how fast the vessel is already turning, so a big turn damps itself. A linear plant has no such term, and the four normalised curves would lie exactly on top of one another.
- **Why the left panel is the proof rather than the right.** Normalisation is what makes the nonlinearity visible: after dividing by the step, any linear system must give one curve. These four are visibly different — the 5° response overshoots to 1.12 while the 120° one never exceeds 1.00. **Superposition fails, and that is the definition of a nonlinear plant.**
- **What changes with the situation.** The 60° and 120° runs share a peak rate of 19.505 deg/s because both saturate the propellers; past that point the vessel turns as fast as it can and the damping multiplier stops rising too, which is why the orange curve flattens. The practical consequence is that the design of §3-3 is a **small-signal** result: honest for small corrections, and conservative — never optimistic — for large ones.

Summary

Week Summary

Step	What was done	How it was verified
1	Identified the heading loop as type 1	steady-state error $\leq 8 \times 10^{-3}$ deg at $K_p = 30$, 100, 300
2	Derived ω_n and ζ from the yaw equation	$K_p = 100$, $K_d = 74.90$ gives the requested $\zeta = 0.900$
3	Showed K_d damping this axis	overshoot $11.74 \to 4.10 \to -0.02$ per cent as K_d rises, the reverse of Week 2
4	Implemented and removed the wrap	20° of turn against 340°, for the same commanded heading
5	Measured the nonlinear yaw damping	overshoot $11.74 \to 0.10$ per cent as the step grows; damping multiplier $1.67 \to 4.40$

Progress Check

Theory

- ☐ Able to state why the heading loop is type 1 without computing anything
- ☐ Able to obtain K_p and K_d from a required ω_n and ζ
- ☐ Able to explain, from the equation of motion, why K_d damps here and did not in Week 2
- ☐ Able to write ssa and say where in the model it belongs

Laboratory

- ☐ `W03_0_setup` printed $\zeta = 0.9000$ and $\omega_n = 1.5312$
- ☐ Sections C, D, E and F were run in that order and together wrote four result PNG files into `W03_simulink/img/`
- ☐ The `use_ssa = 0` run was watched in the live view to the end

Recorded observations

- ☐ The steady-state error at three gains, with the averaging window stated
- ☐ Predicted and measured overshoot for all four K_d, and the size of the gap
- ☐ Total turn with and without the wrap
- ☐ Peak $|r|$ for each step size, and the damping multiplier it implies

---

Assignment 3

- **Due**: before the Week 4 session
- **Submit**: the modified `W03_0_setup.m`, a derivation, the numbers requested below, and one figure

① Requirements

1. Derive ω_n and ζ for the heading loop from the yaw equation of motion, showing each step. Then determine, by hand, the pair (K_p, K_d) that gives $\omega_n = 1.0$ rad/s and $\zeta = 0.7$.
2. Determine the largest K_p for which the hull's own damping alone, with $K_d = 0$, still gives $\zeta \geq 0.5$.

3. Determine, by hand, the yaw moment τ_N demanded at the instant a 90° step is applied with the design of ①.1, and state whether it lies inside the attainable set $|N| \leq 73.62$ N·m of Appendix A1.

② Verification — mandatory

> **★ A claim is not a result. Numbers are required, with the tolerance band and the averaging window stated**

1. Run the design of ①.1 with a 5° step and report the measured overshoot and 2% settling time. Compare with the prediction and state the gap.
2. Repeat ②.1 with a 90° step. Report both, and explain the difference using $1 + 10|r|$ at the measured peak rate.
3. Run the design of ①.2 and report the measured overshoot. State whether the hull alone was enough.
4. Command $+179^\circ$, let it settle, then command -179°. Report the total turn with `use_ssa = 1` and with `use_ssa = 0`, and the time each takes to settle. Produce **one figure** of heading against time for both.

③ Analysis (6–10 lines)

- Item ②.2 shows a prediction that is wrong in a specific direction. State the direction, explain it from $N_h = N_r(1 + 10|r|)r$, and say whether a controller designed with the small-signal equations is therefore **safe** or **unsafe** for large steps. Then state one situation in which the same nonlinearity would be a problem rather than a help.

Grading

Criterion	Weight
Derivation in ① is complete and correct	20%
Verification performed and numbers reported with bands and windows stated	40%
The figure in ②.4 is correct and readable	15%
Analysis in ③ identifies the direction of the error and its consequence	25%

Troubleshooting

Symptom	Cause	Fix
The vessel turns almost all the way round for a small heading change	`use_ssa = 0`	set `use_ssa = 1`; see §3-5
The heading plot passes 360° and keeps rising	ψ is not wrapped, by design	expected. Only the error is wrapped
The measured overshoot is far below the prediction	the step is large enough for $N_h = N_r(1 + 10|r|)r$ to matter	expected; see §F. Use a small step to test a linear design
Adding K_d makes the response slower but not less damped	K_d is already past critical, $\zeta > 1$	reduce K_d; $\zeta = 1$ is at $K_d = 87.96$ for $K_p = 100$
Peak yaw rate is the same for two different step sizes	the propellers are saturating	expected; the vessel turns as fast as it can
The vessel drifts sideways while turning	the crab angle of Week 1	expected, and the subject of Week 4

References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed. §12.2.7 (PID heading autopilot) and §15.3 (course and heading autopilots).
- MSS toolbox, `Tools/MSS/GNC/ssa.m` — the wrap used throughout MSS.
- MSS toolbox, `Tools/MSS/SIMULINK/mssSimulinkDemos/demoOtterUSVHeadingControl.slx` — the demonstration whose layout every model in this course follows.

Course files

- `W03_simulink/W03_1_build_heading.m` — the model generator
- `W03_simulink/W03_0_setup.m` — the parameters, including the design equations
- `W03_simulink/W03_C_proportional_only.m` · `W03_D_derivative_action.m` · `W03_E_the_wrap.m` · `W03_F_big_turns_overshoot_less.m` — one script per laboratory section
- `W03_simulink/W03_vars.m` · `W03_read.m` — the same numbers as a struct, and the log with the angles in degrees
- `W03_simulink/W03_plot.m` — the summary figure, called by both the runner and the model's `StopFcn`
- `_tools/gnc_chain.m`, `_tools/add_subsys.m`, `_tools/add_measurement.m` — the standard layout

Next Week

- **Week 4 — Waypoint Following and LOS Guidance**
- This week was told what heading to hold; a human typed the number into a step block. Week 4 asks where that number comes from, and the answer is a list of waypoints and a guidance law.
- The cross-track and along-track errors from one rotation, the line-of-sight law, and the two laws — integral and adaptive — that survive an ocean current.
- Preparation: §3-4 of this week, on the crab angle. Week 4 opens by quantifying exactly what it costs.